Creative Engineering Design

Dr. Sarvar Abdullaev

Course Info

- Lecturer: Dr. Sarvar Abdullaev
- Email: s.abdullaev@inha.uz
- Office hours:
  - Monday 13:00 – 14:00 in Room B411
  - Friday 13:00 – 14:00 in Room B411
- Telegram Channel:
  - **t.me/iut_ced2023**
- Course format:
  - 2 lectures every week
  - 1 homework assignment every week

What is this course about?

creative

engineering

- Problems and their non-conventional solutions from various scientific fields
- Nature-inspired solutions to some engineering and computational problems
- Abstract idea models for deeper understanding of certain natural or social phenomenon
- Interdisciplinary and system thinking approaches in solving social issues

design

- Iterative design processes
- User Experience (UX) Research
- User Interface (UI) Design
- Conceptual Product Design
- Prototyping and Modelling Products
- Experiments Design and Testing
- Measuring and Reflecting on Process

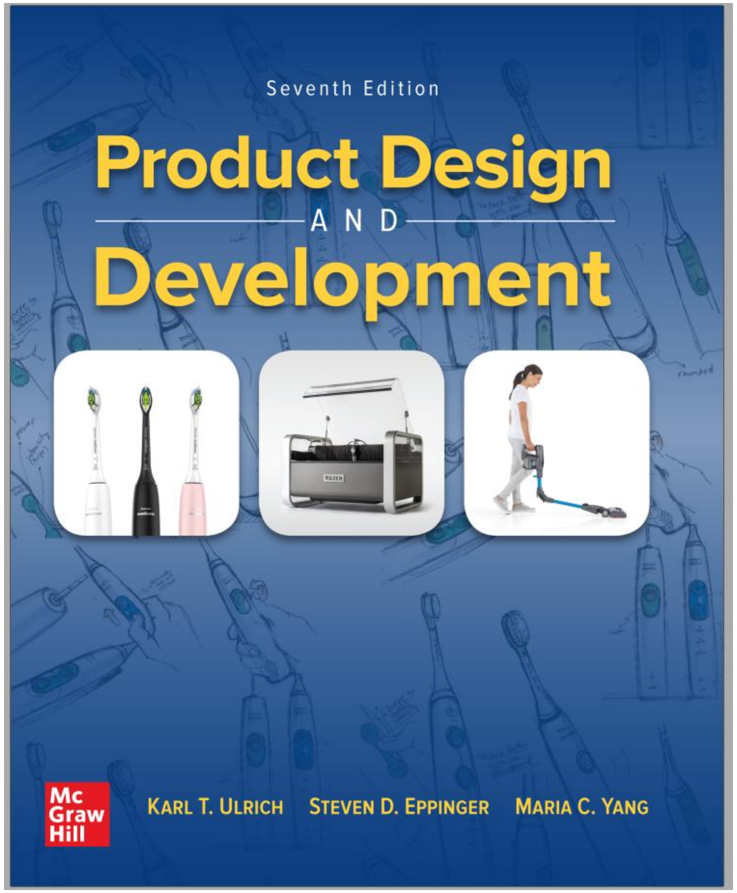
Course Outcomes

- Define the problem clearly using engineering terms from the collected voice of customers
- Generate creative ideas to solve the problem through team activities
  - Thinking divergently and convergently
  - Organizing and documenting team meetings
  - Writing reports on accomplished tasks
- Evaluate and refine the ideas to obtain the final solution
- Implement the idea to a real operating product
  - Product design and architecture
  - Prototyping and user experience
- Present the design result clearly
  - Oral presentation and technical documentation
- Communicate effectively with other team members, advisors, and general audience through the whole process

Main Textbook

- K. Ulrich, S. Eppinger and M. Yang, *Product Design and Development (7th ed),* McGraw Hill, USA - 2020

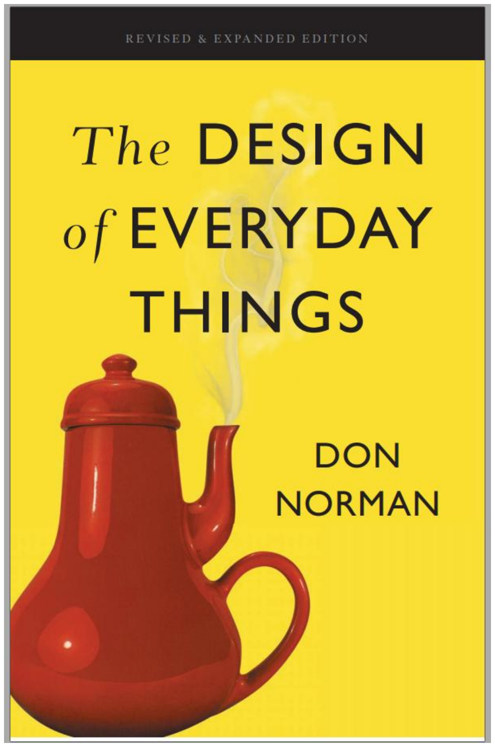
PD&D book



Additional References

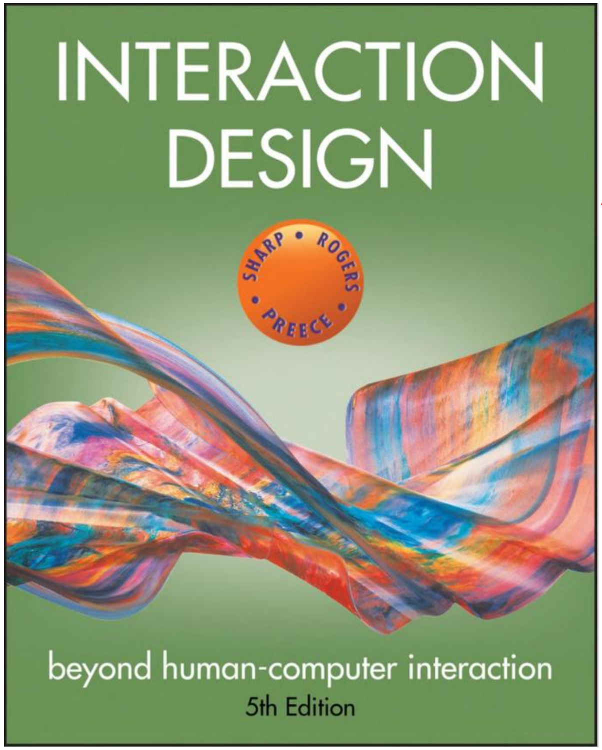
- Don Norman, *The Design of Everyday Things,* Basic Books, US - 2013

Norman's book



- Helen Sharp, Yvonne Rogers and Jennifer Preece. *Interaction design: beyond human-computer interaction,* John Wiley & Sons, 2015.

IxD book



Some Useful Online Courses

- Short Online Courses (5-7 weeks long, one lecture per week):
  - Innovation Through Design: Think, Make, Break, Repeat
    - https://www.coursera.org/learn/innovation-through-design
  - Product Design: The Delft Design Approach
    - https://www.edx.org/course/product-design-the-delft-design-approach
- Online Specializations (6-10 months long, 6-7 online courses per specialization)
  - Google UX Design Professional Certificate
    - https://www.coursera.org/professional-certificates/google-ux-design
  - Interaction Design Specialization
    - https://www.coursera.org/specializations/interaction-design

Grading policy

- Individual assignments: **10%**
  - Assignments are given over eClass
  - They are marked for completion and **NOT GRADED.**
- Three Major Assessment Components: 20%+20%+40% = **80%**
  - Two conceptual design projects – group-based:
    - UX/UI prototype of a **digital product** - 20%
    - 3D model of a **physical product** - 20%
  - Product design project – group-based:
    - 3D printed prototype of a **working physical product** - 40%
  - … OR Individual Term Paper – individual-based
    - Putting a known model into a non-conventional use – 40%
- Attendance: **10%**
- No Exams!

Introduction to Creative Engineering Design

Lecture 1

What is engineering?

"Engineering is the development of cost-effective solutions to practical problems, through the application of scientific knowledge."

- "… cost-effective …"
  - Consideration of design trade-offs, especially resource usage
  - Minimize negative impacts (environmental and social cost)
- "… solutions …"
  - Emphasis on building devices and systems
- "… practical problems …"
  - Solving problems that matter to people
  - Improving human life in general through technological advance
- "… application of scientific knowledge …"
  - Systematic application of analytical techniques

Slightly modified from: http://www.cs.toronto.edu/~sme/CSC340F/slides/03-engineering.pdf

Engineering vs. Science

Traditional view

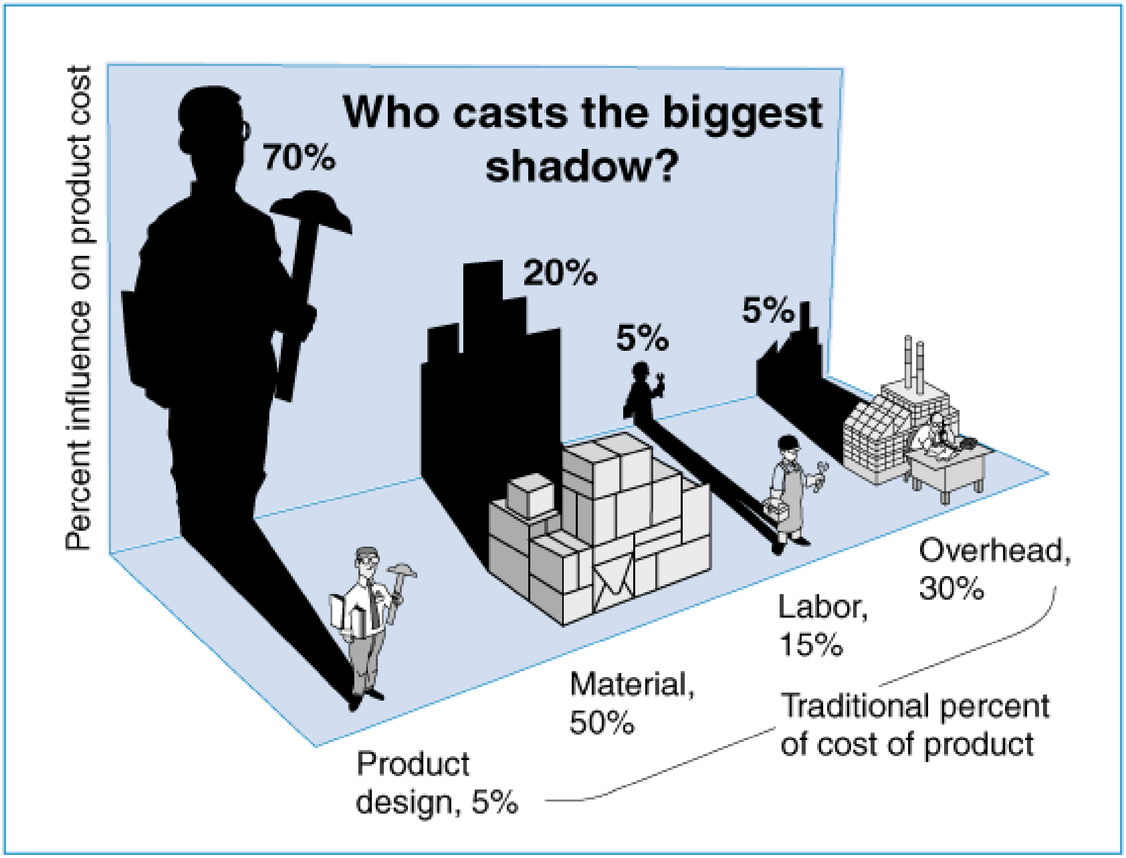
- Scientists
  - create knowledge
  - study the world as it is
  - are trained in scientific method
  - use explicit knowledge
  - are thinkers
- Engineers
  - apply knowledge
  - seek to change the world
  - are trained in engineering design
  - use tacit knowledge
  - are doers

More realistic view

- Scientists
  - create knowledge
  - are problem-driven
  - seek to understand and explain
  - design experiments to test theories
- Engineers
  - create knowledge
  - are problem-driven
  - Seek to understand and explain
  - Design devices to test theories

From: http://www.cs.toronto.edu/~sme/CSC340F/slides/03-engineering.pdf

Is engineering design important?



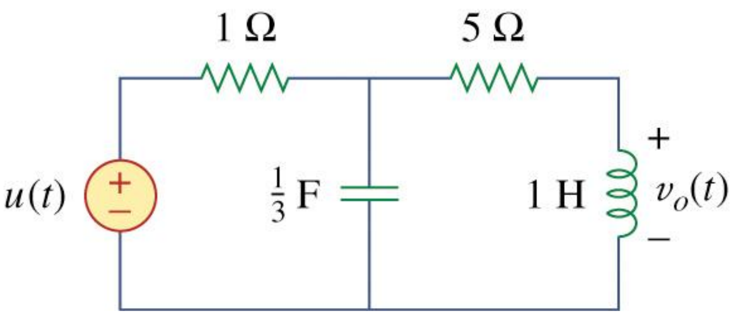
- Yes!
- Percent influence of the engineering design on product cost is 70%
- 85% of problems in new products can be attributed to bad engineering design
  - Costs are too high
  - Does not operate as intended
  - Low reliability
  - Time needed to enter the market is too long

Development efforts of different products

	Belle-V Ice Cream Scoop	AvaTech Avalanche Probe	iRobot Roomba Vacuum Cleaner	Tesla Model S Automobile	Boeing 787 Aircraft
Annual production volume	10,000 units/year	1,000 units/year	2,000,000 units/year	50,000 units/year	120 units/year
Sales lifetime	10 years	3 years	2 years	5 years	40 years
Sales price	$40	$2,250	$500	$80,000	$250 million
Number of unique parts (part numbers)	2 parts	175 parts	1,000 parts	10,000 parts	130,000 parts
Development time	1 year	2 years	2 years	4 years	7 years
Internal development team (peak size)	4 people	6 people	100 people	1,000 people	7,000 people
External development team (peak size)	2 people	12 people	100 people	1,000 people	10,000 people
Development cost	$100,000	$1 million	$50 million	$500 million	$15 billion
Production investment	$20,000	$250,000	$10 million	$500 million	$15 billion

Engineering design:
Deals with open-ended problem.

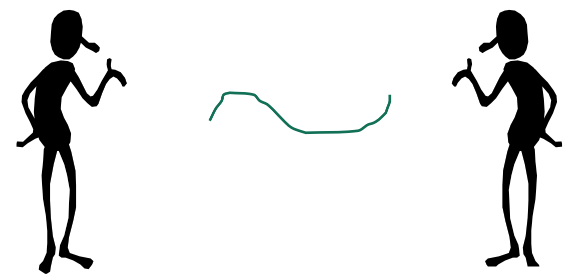
Ex) Find voltage v(t) for a given circuit



$$v(t) = \frac{3}{\sqrt{2}} e^{-4t} \sin\sqrt{2}t, \quad t \geq 0$$

Only one solution

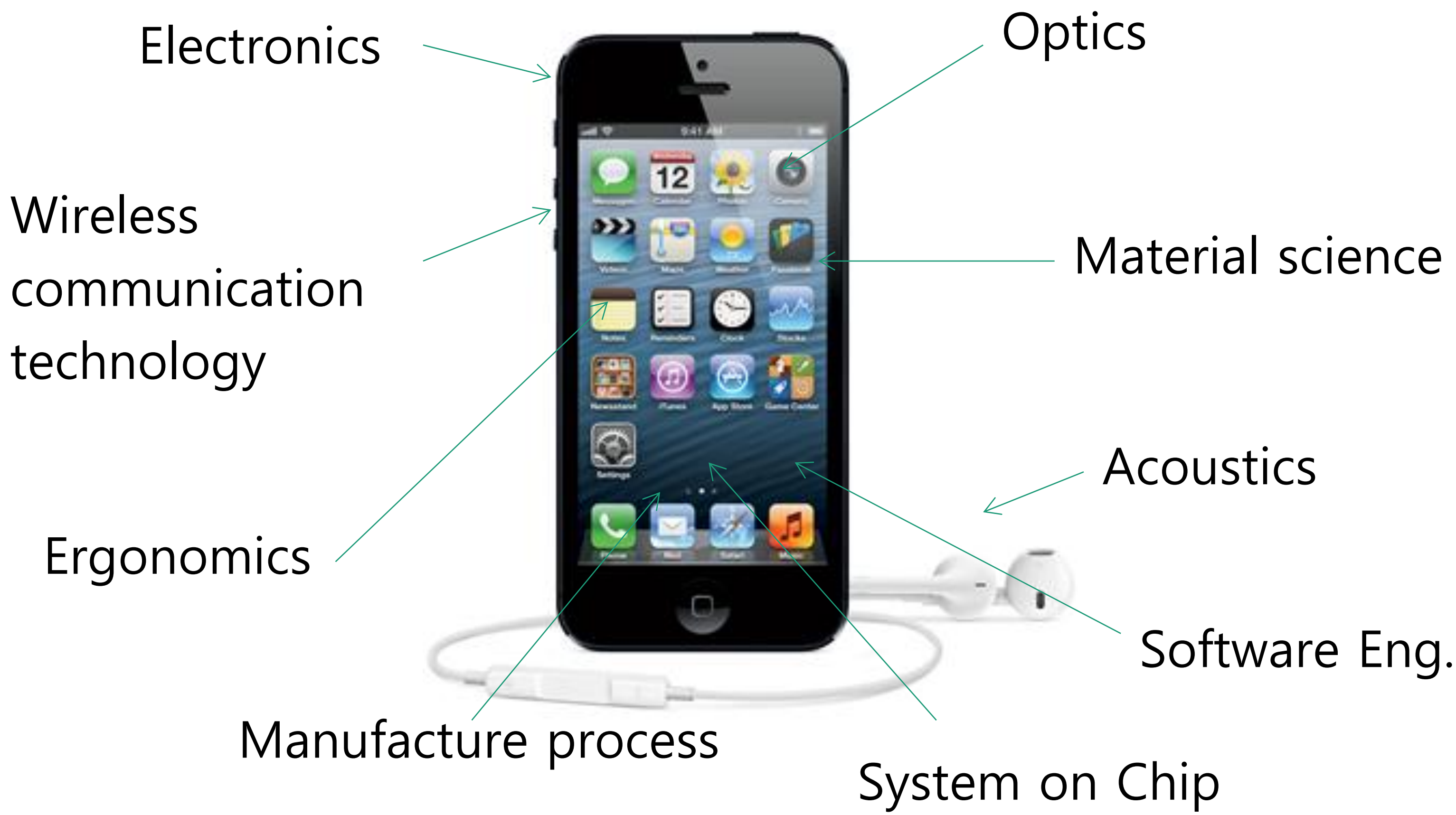
Ex) Design a device that enables people to talk remotely



Various solutions possible

Engineering design:

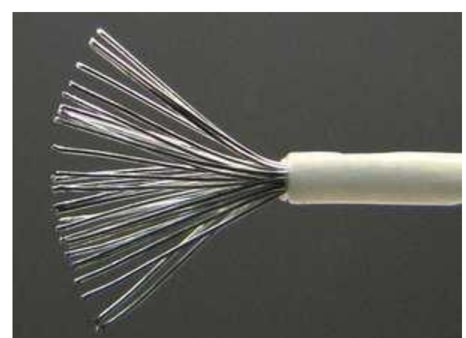
Requires knowledge and skill from various fields



Engineering design:
Requires cost consideration

Position	Name	Symbol	Conductivity S/m*10^6
1	Silver	Ag	62,89
2	Copper	Cu	59,77
3	Gold	Au	42,55
4	Aluminium	Al	37,66
5	Brass, 10% zinc	CuZn10	35,86
6	Rhodium	Rh	22,17
7	Zinc	Zn	[illegible]

Silver cable for every electric system?

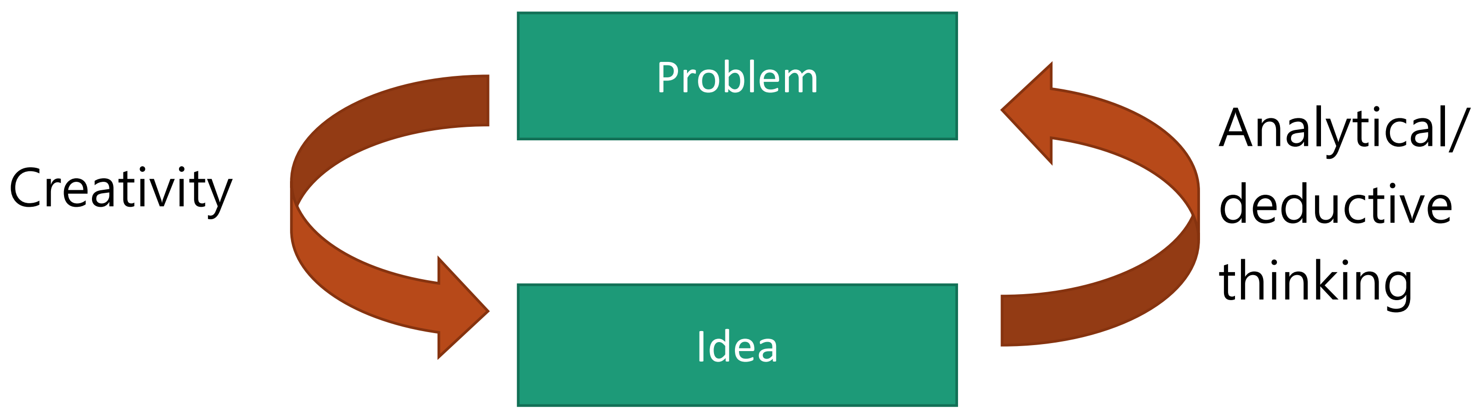


Engineering design
Requires effective technical communication

- Communication within a design group
- Communication with other people (manager, people in other departments, general audience, peer researcher, investor, …)
- Documentation (Design report, Meeting minutes, Project proposal, Technical drawing, etc …)
- Oral presentation

Engineering design
Requires creativity

- Creative ideas to solve engineering problems
- But also requires analytical, deductive thinking
- Brain storming, scketching, …



What is interaction design?

"Designing interactive products to support the way people communicate and interact in their everyday and working lives."

Sharp, Rogers, and Preece (2019)

"The design of spaces for human communication and interaction."

Winograd (1997)

Goals of interaction design

Develop usable products

- Usability means easy to learn, effective to use, and provides an enjoyable experience

Involve users in the design process

Which kind of design?

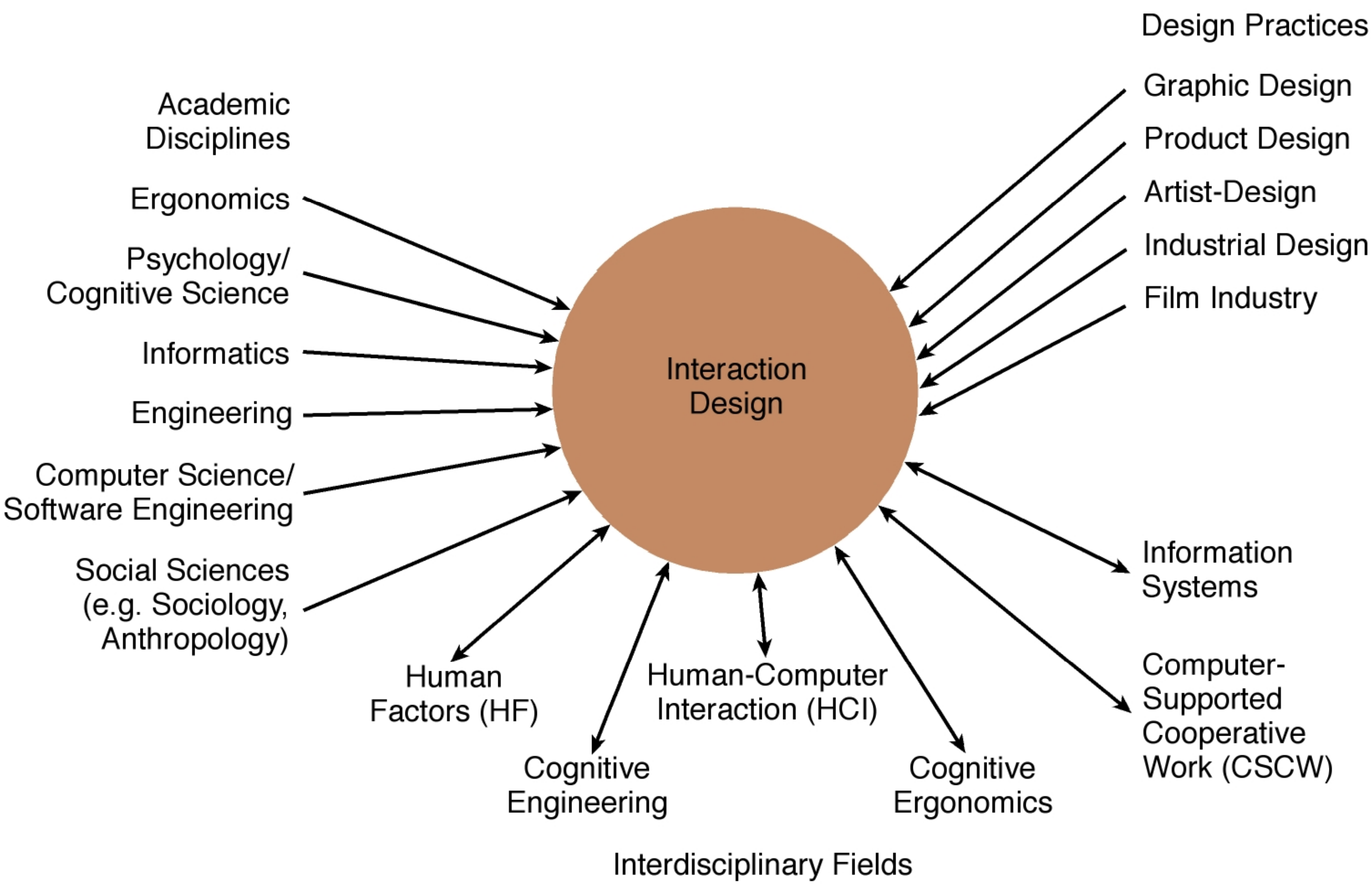
Number of other terms used emphasizing what is being designed, for example:

- User interface design, software design, user-centered design, product design, web design, experience design (UX)

Interaction design is the umbrella term covering all of these aspects:

- Fundamental to all disciplines, fields, and approaches concerned with researching and designing computer-based systems for people

Interaction design



Elevator controls

Elevator controls and labels on the bottom row all look the same, so it is easy to push a label by mistake instead of a control button.

www.baddesigns.com

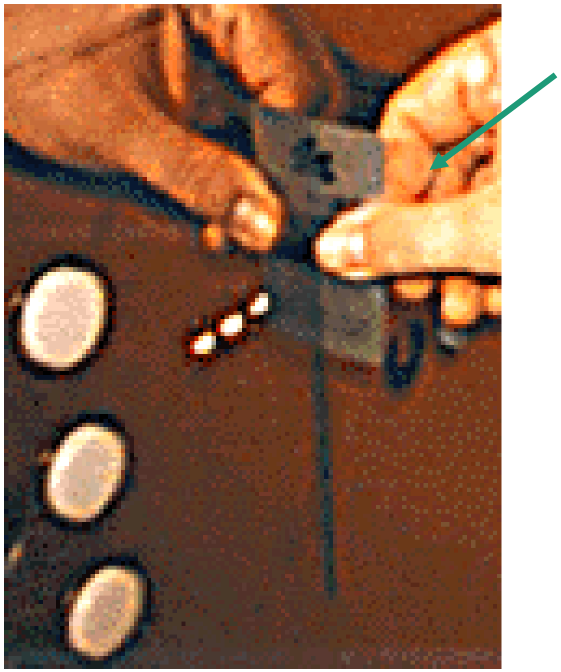
People do not make same mistake for the labels and buttons on the top row. Why not?

Visibility - poor interface

www.baddesigns.com

- This is a control panel for an elevator
- How does it work?
- Push a button for the floor you want?
- Nothing happens. Push any other button? Still nothing. What do you need to do?
- It is not visible as to what to do!

Visibility - Improving on a poor interface



www.baddesigns.com

…with this elevator, you need to insert your room card in the slot by the buttons to get the elevator to work!

How would you make this action more visible?

- Make the card reader more obvious
- Provide an auditory message that says what to do (which language?)
- Provide a big label next to the card reader that flashes when someone enters
- Make relevant parts visible
- Make what has to be done obvious

Why is this vending machine so bad?

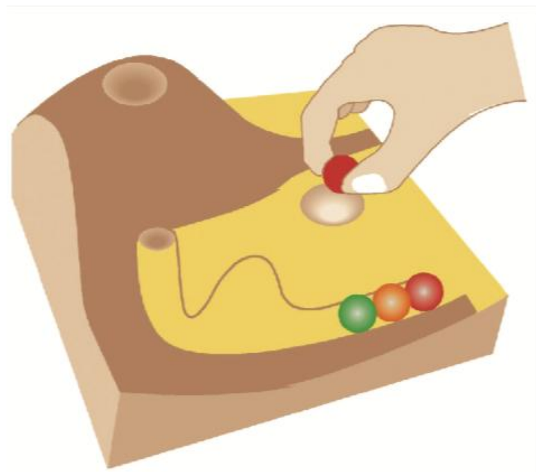
www.baddesigns.com

- Need to push button first to activate reader
- Normally insert bill first before making selection
- Contravenes well known convention

Telephone with Voice-mail recorder

- Red-light blinking. What does it mean?
- Oh, I have voice-mail. What should I do next?
- How can I browse my voice-mail? Oh, these small arrows. Password? What is my password?
- …

Better voice-mail design concept



- Marble answering machine (Bishop, 1995)
- Based on how everyday objects behave
- Easy, intuitive, and a pleasure to use
- Only requires one-step actions to perform core tasks

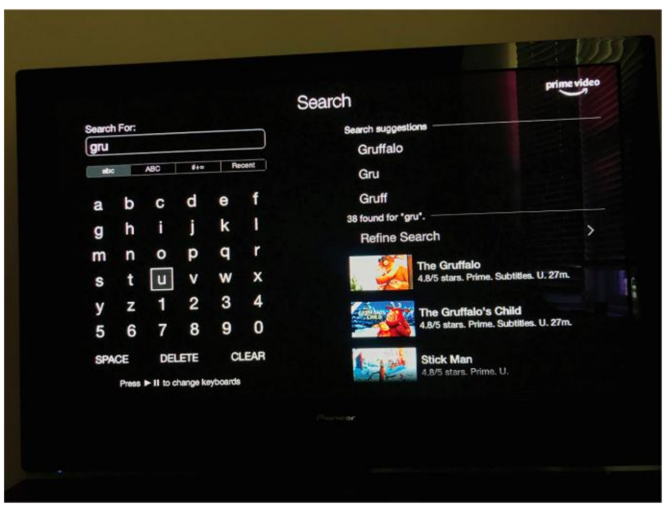
TV Remotes

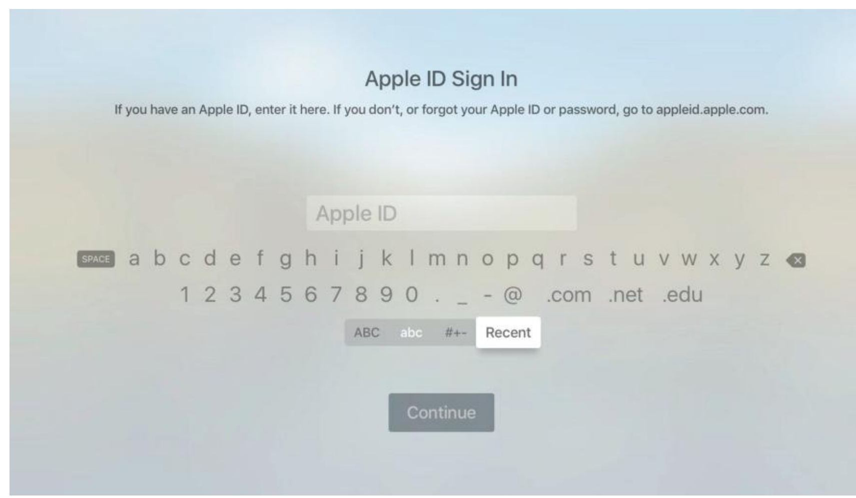
- Difficult to locate buttons even for simple tasks
- Difficult to understand what buttons mean
- Most of the functionality is not used
- Sometimes too clunky to hold

What user sees in the screen.

Which is the best way to interact with a smart TV? Why?

- Pecking using a grid keyboard via a remote control
- Swiping across two alphanumeric rows using a touchpad on a remote control
- Voice control using remote or smart speaker





Improved TV remote

Why is the Xiomi remote much better designed than standard remote controls?

- Smaller in size
- Less cluttered buttons
- Better spatial mapping of buttons
- Familiar labels and icons on buttons
- Speech recognition

What is UX?

How users perceive a product, such as whether a smartwatch is seen as sleek or chunky, and their emotional reaction to it, such as whether people have a positive experience when using it.

(Hornbæk and Hertzum, 2017)

Hassenzahl’s (2010) model of the user experience

- Pragmatic: how simple, practical, and obvious it is for the user to achieve their goals
- Hedonic: how evocative and stimulating the interaction is to users

Core characteristics of interaction design

- Users should be involved throughout the development of the project

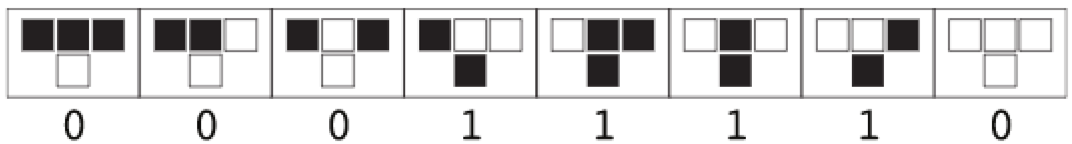
- Specific usability and user experience goals need to be identified, clearly documented, and agreed to at the beginning of the project

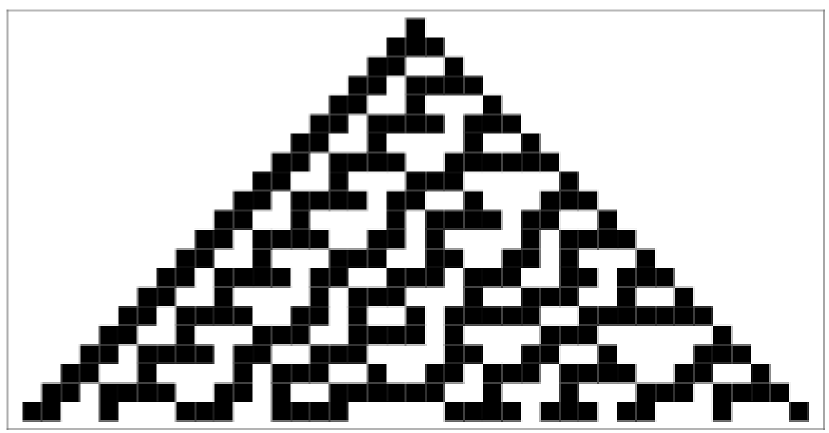
- Iteration is needed through the core activities

Cellular Automata

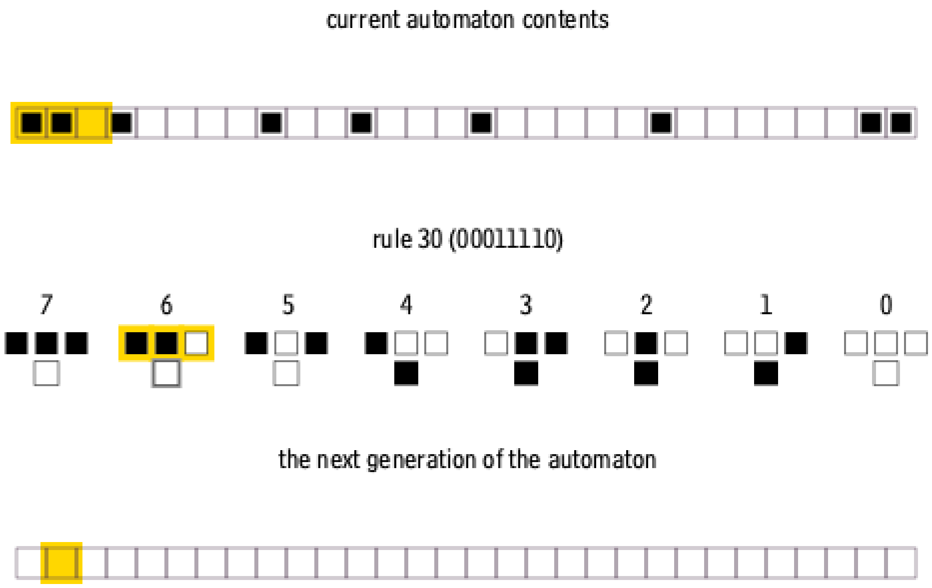
Engineering Idea #1

1D Elementary Cellular Automaton

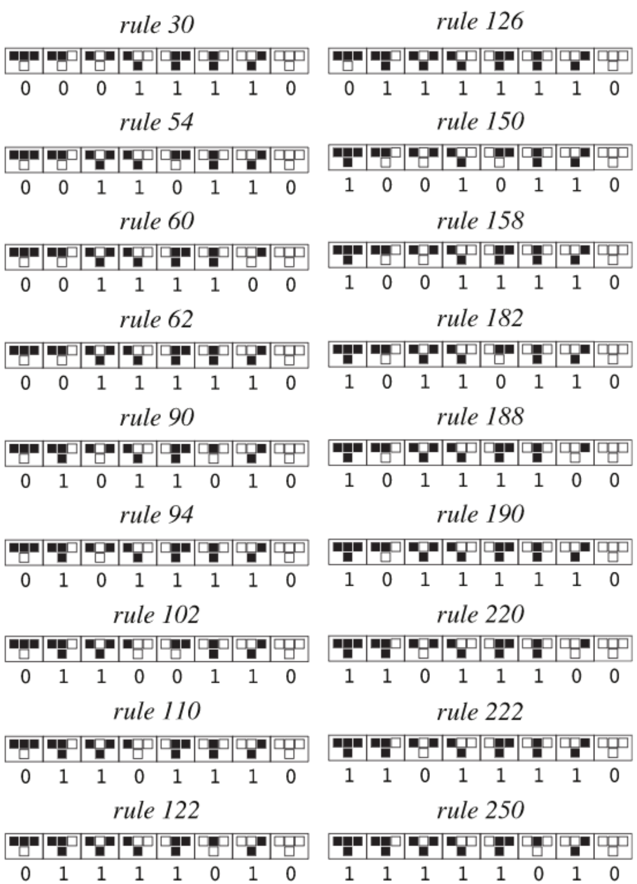


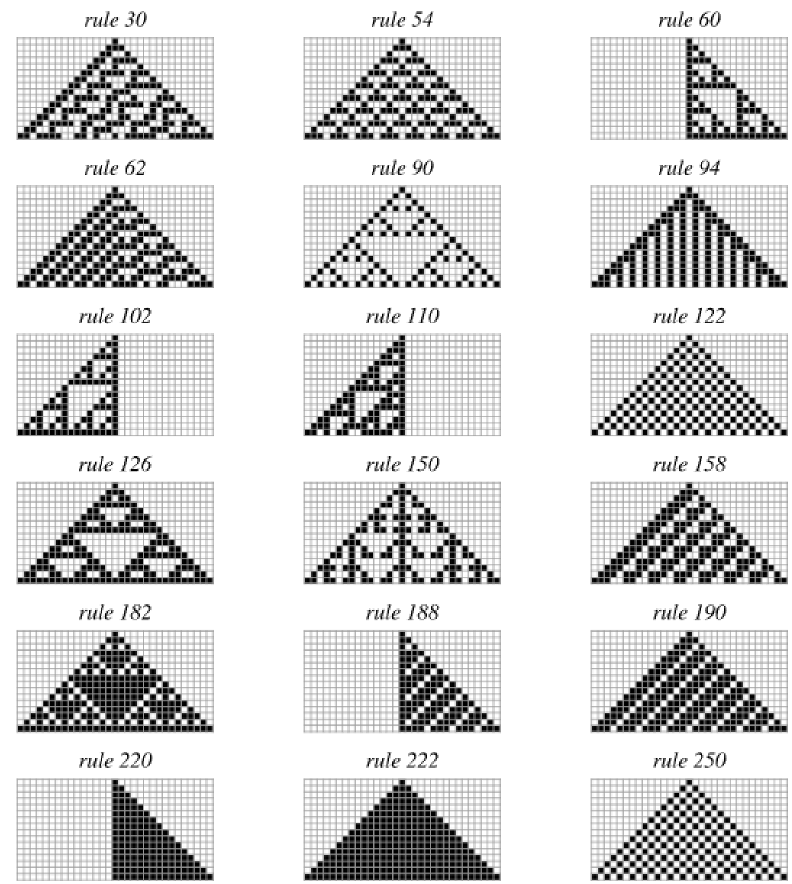


1D Elementary Cellular Automaton



1D Elementary Cellular Automaton





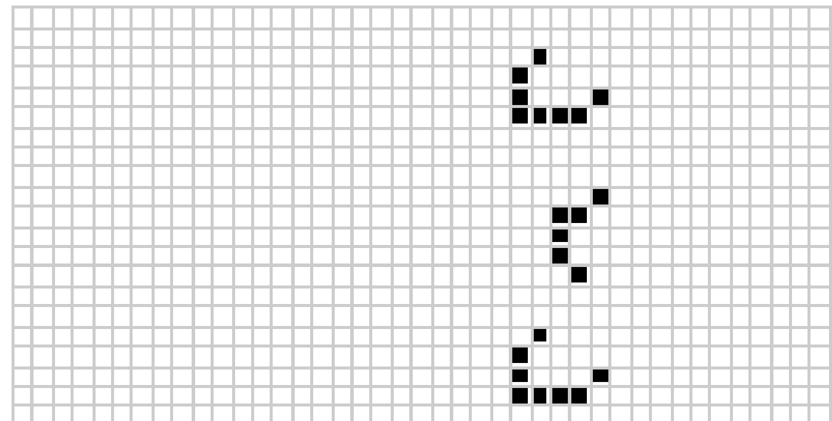
Mathematical Properties

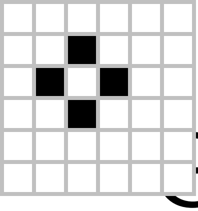
$$a_i(t) = f\,(a_{i-1}(t-1),\ a_i(t-1),\ a_{i+1}(t-1)).$$

$$f_{30}(p, q, r) = \mathrm{Xor}[p, \mathrm{Or}[q, r]]$$
$$f_{90}(p, q, r) = \mathrm{Xor}[p, r]$$
$$f_{110}(p, q, r) = \mathrm{Xor}[\mathrm{Or}[p, q], \mathrm{And}[p, q, r]]$$
$$f_{250}(p, q, r) = \mathrm{Or}[p, r]$$
$$f_{254}(p, q, r) = \mathrm{Or}[p, q, r]$$

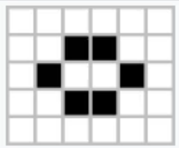
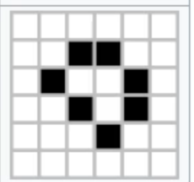
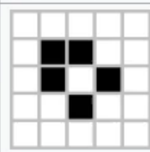
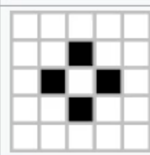
Game of Life – 2D totalistic CA

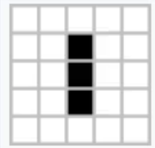
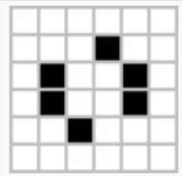
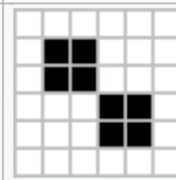
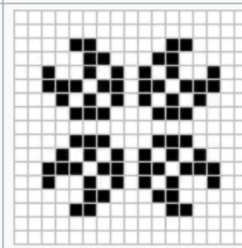
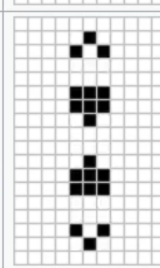
- Any live cell with two or three live neighbours survives.
- Any dead cell with three live neighbours becomes a live cell.
- All other live cells die in the next generation. Similarly, all other dead cells stay dead.

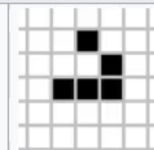
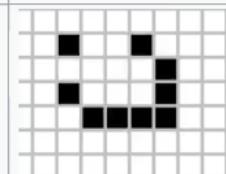
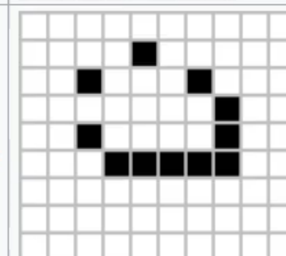
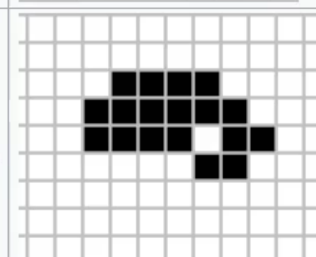




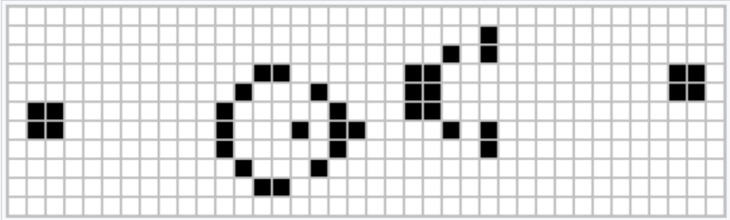
Game of Life - Zoo

Still lifes	
Block	
Bee-hive	
Loaf	
Boat	
Tub	

Oscillators	
Blinker (period 2)	
Toad (period 2)	
Beacon (period 2)	
Pulsar (period 3)	
Penta-decathlon (period 15)	

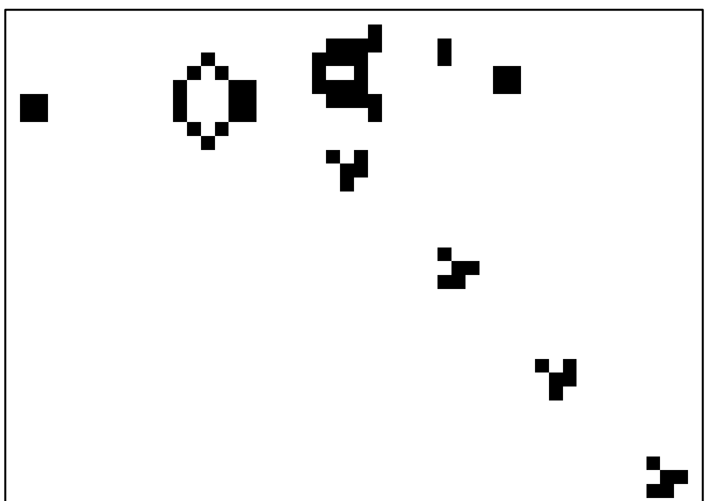
Spaceships	
Glider	
Light-weight spaceship (LWSS)	
Middle-weight spaceship (MWSS)	
Heavy-weight spaceship (HWSS)	

Game of Life – Generating/Blocking Signals

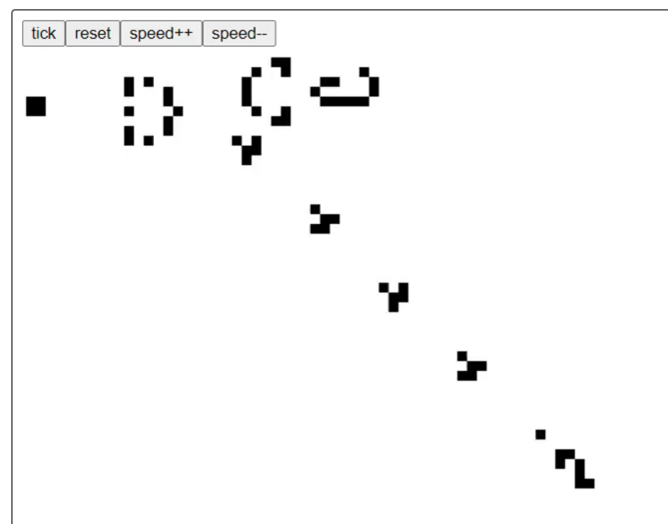

Gosper glider gun

Gospel's glider gun

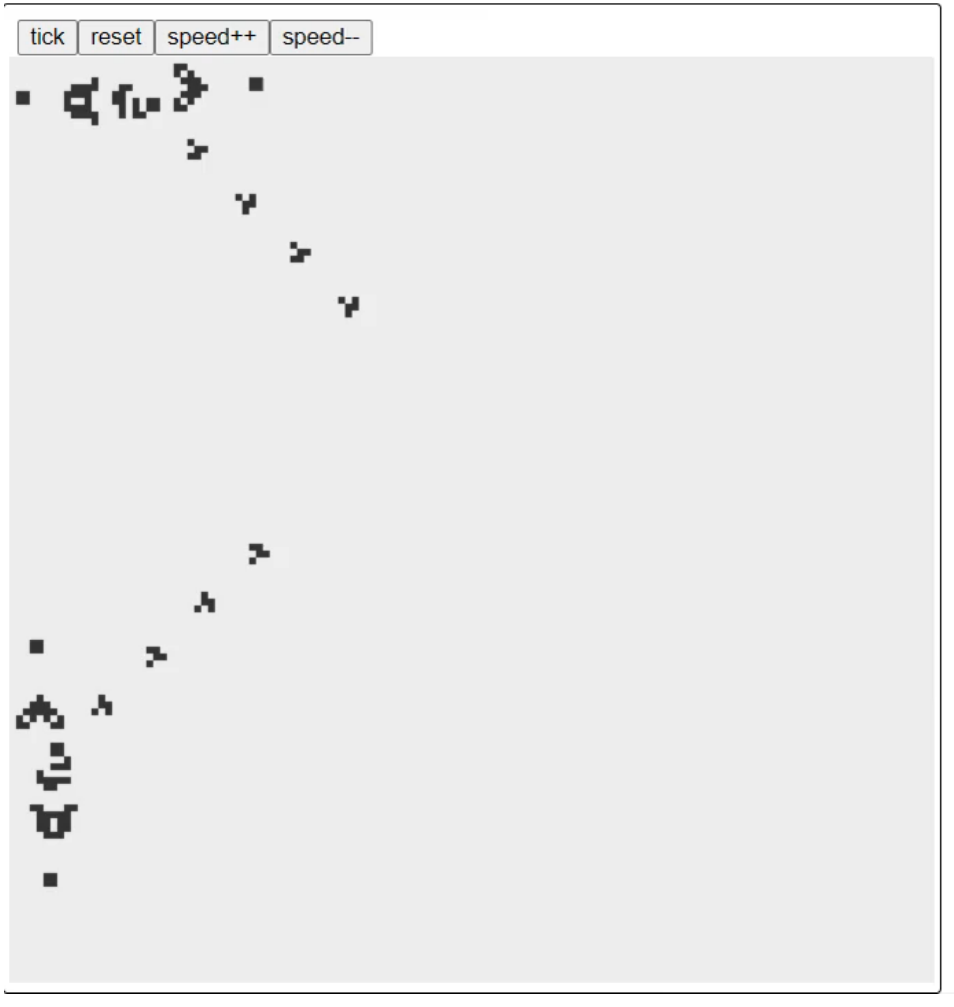
Generating signals



Blocking signals

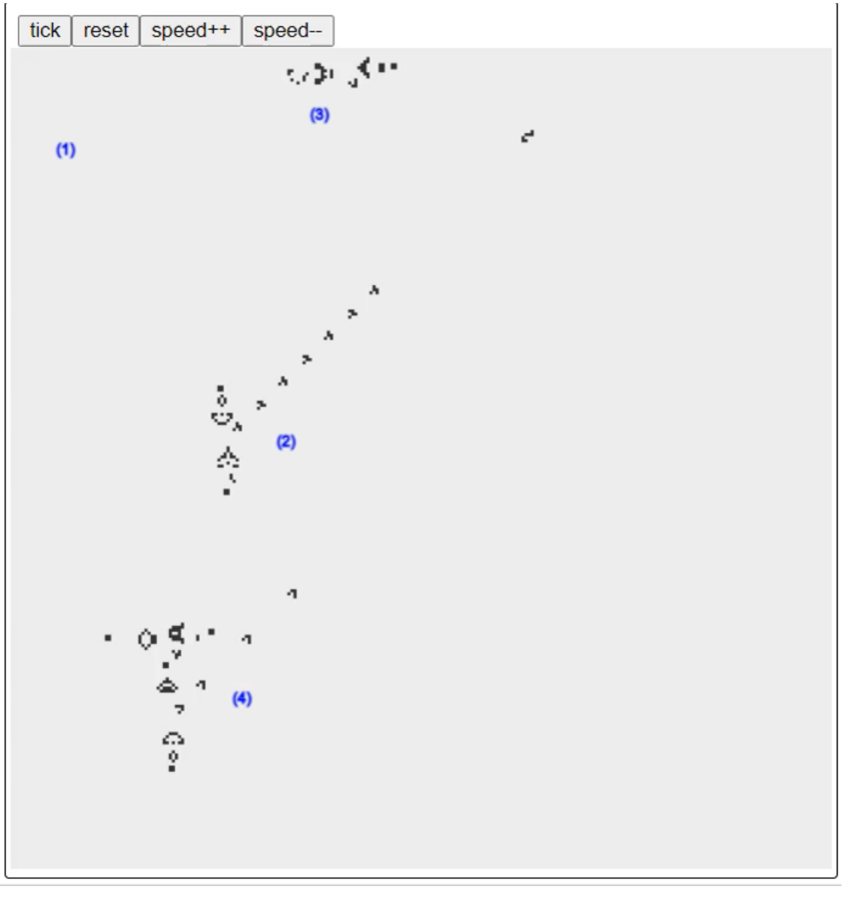


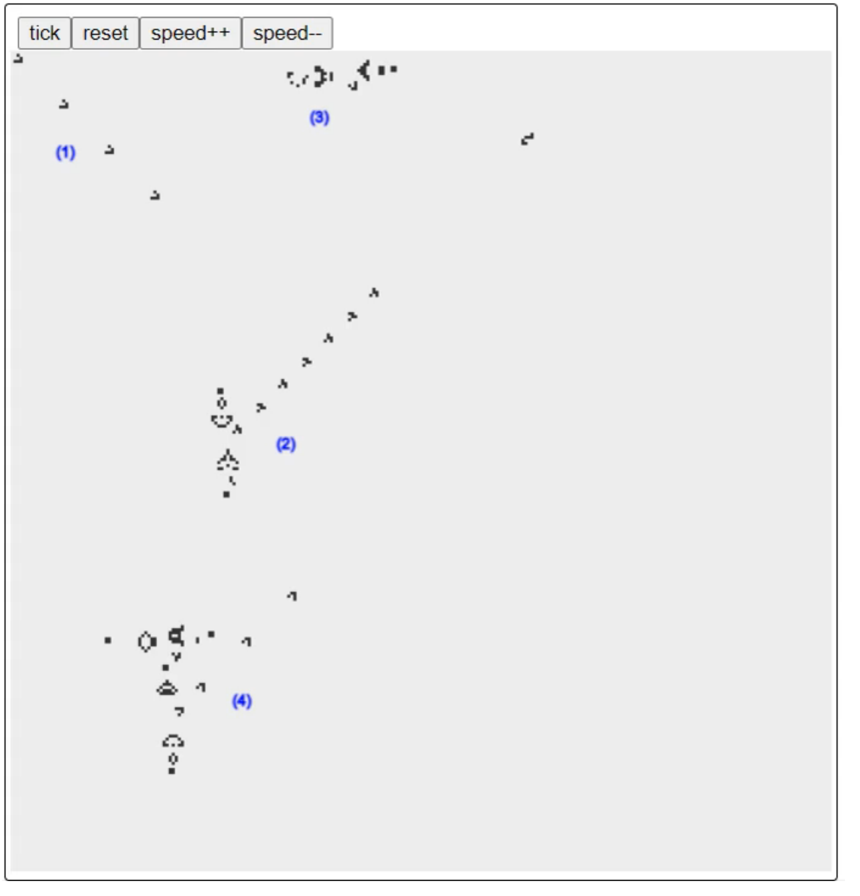
Game of Life - Collisions



Game of Life – NOT gate

- (1) is false, (4) is true
- (1) is true, (4) is false



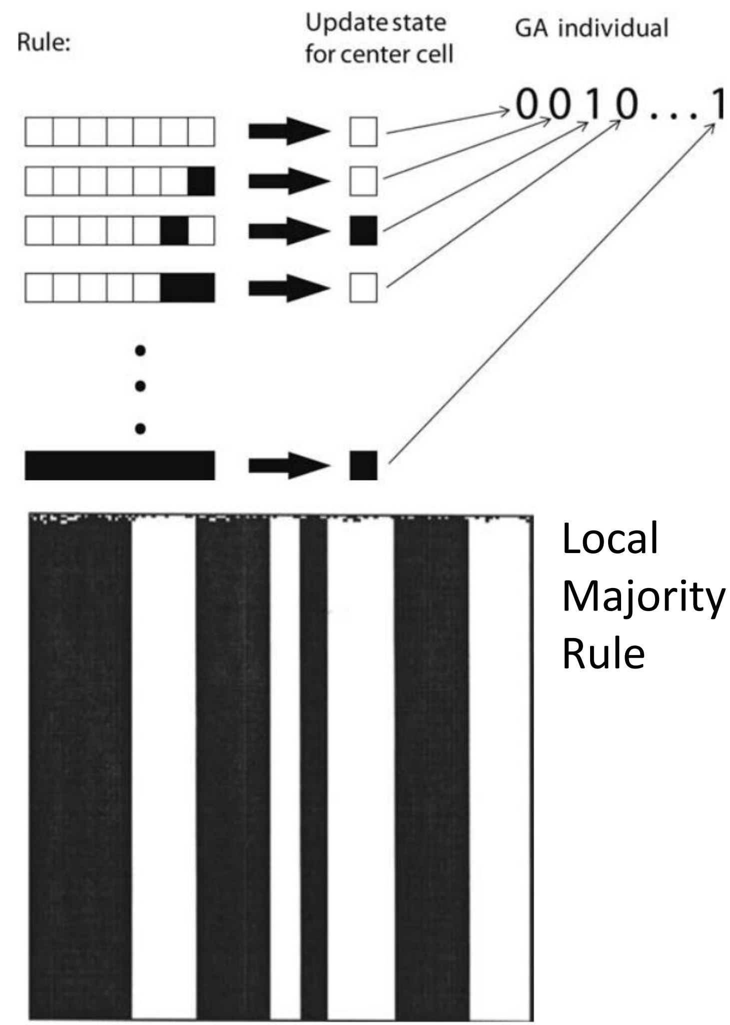


Game of Life – AND gate

- See here:
- https://nicholas.carlini.com/writing/2020/digital-logic-game-of-life.html

Computing with Particles

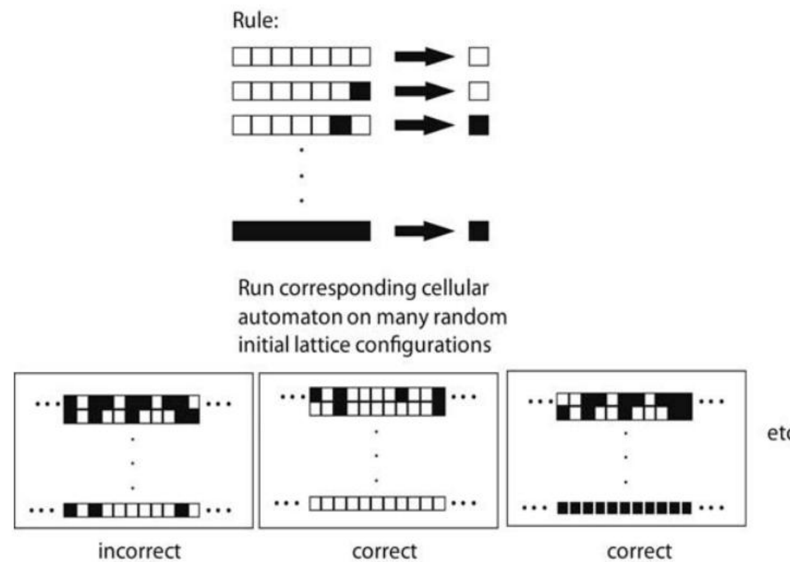
- Every cell can see itself, and 3 of its left and right neighbours.
- There 128 possible configurations of cells.
- For each of this configurations, cell has to decide to turn on/off.
- Can cells collectively decide who is in majority? On cells or off cells?
- If majority is off, after 200 iterations all are off, and vice versa.



Local Majority Rule

Cells trained by Genetic Algorithms

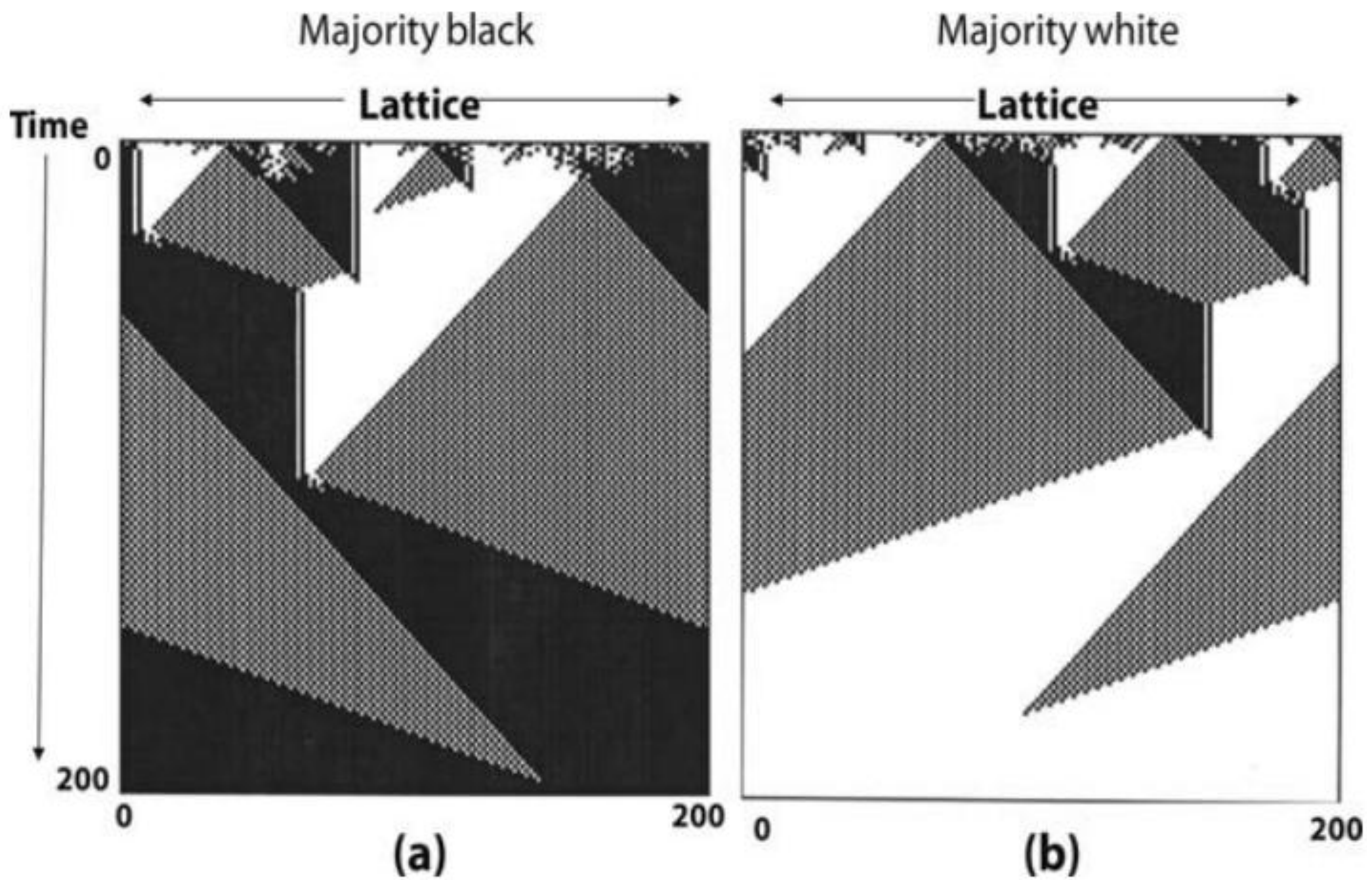
- Genes – rules for each state
- Each cell has the same gene
- Fitness of given gene is the ratio of times it end up correctly.



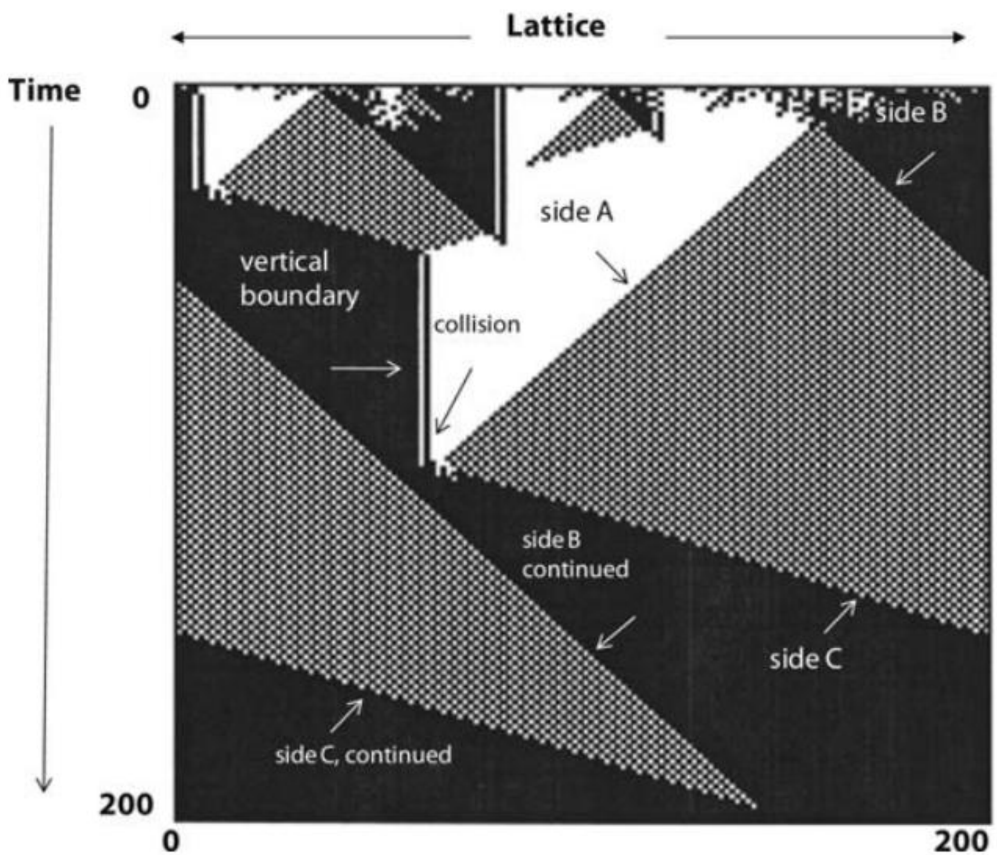
Best gene

0000010100000011000010101100001110000001110000001000001010101010111
01100100011101110000010100000000101111101111111111011011101111111

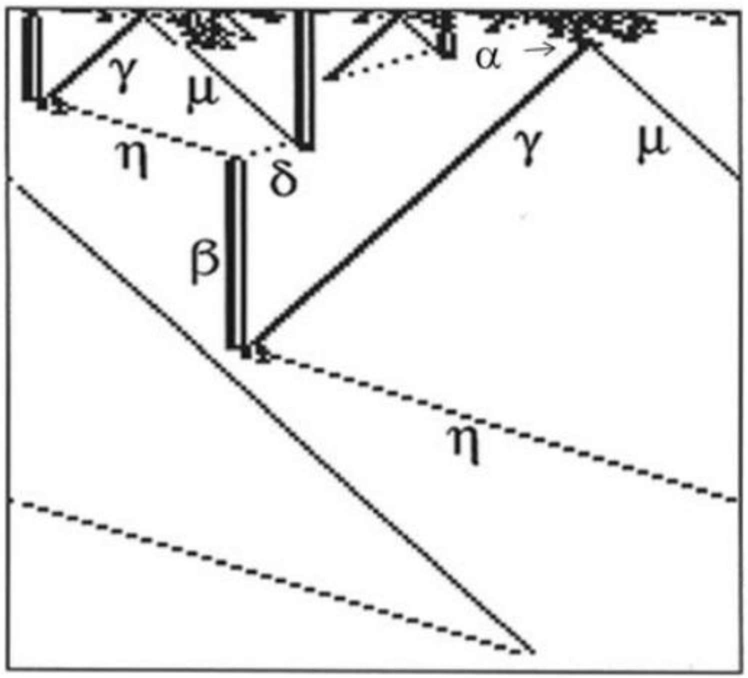
Space-Time Diagram



Gaining Insights



Formalizing Laws



- 6 different particles:
  - γ (gamma – white & checkboard),
  - μ (mu – checkboard & black),
  - η (eta – black & checkboard),
  - δ (delta – checkboard & white),
  - β (beta – black & white),
  - α (alpha, a short-lived particle that decays into γ and μ)
- 5 types of particle collisions:
  - $\beta + \gamma = \eta$
  - $\mu + \beta = \delta$
  - $\eta + \delta = \beta$
  - $\eta + \mu =$ (black)
  - $\gamma + \delta =$ (white)

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